GEOGRAPHY CH – 5 LANDFORMS OF THE EARTH

1. Introduction to Landforms & Internal Heat 🌍🔥

Definitions:

- **Landform:** A natural feature of the earth's surface. 🏞️

Key Points:

- The earth is made up of a series of concentric rock layers: crust, mantle, and core. 🧅
- The intense heat in the earth's interior does not travel to the surface because rocks are bad conductors of heat. 🛡️
- This internal heat melts the rocks, keeping the asthenosphere (underneath the crust) in a semi-molten state, which leads to conditions that cause landform development. 🍰

The "Why?" Section ❓

- **Why are there forces moving the plates?** The internal Endogenic forces move the plates due to a *convection flow* caused by *radioactive decay and residual heat.* ☢️🔥 *(Extracted from your handwritten flowchart!)*

Things to Remember 🧠

Feature	Detail
Syllabus Targets	Mountains (fold, residual, block), Plateaus (intermontane, volcanic), Plains (structural, depositional). 🎯

2. Plate Tectonics & Boundaries 🗺️💥

Definitions:

- **Lithosphere:** The solid outer part of the earth, which consists of the *crust + upper mantle.* 🧩 *(Extracted from your handwritten note!)*
- **Lithospheric Slab (Plate):** A broken piece of the lithosphere that contains *both ocean and continent.* 🌊🌍 *(Extracted from your handwritten margin!)*

- **Plate Tectonics:** The process where the lithosphere is broken into several small and big plates that float and move very slowly horizontally on the semi-molten asthenosphere. 🐢

Key Points:

- Plates push, collide, or slide past each other, creating disturbances in sedimentary rock deposits lying between them. 💥
- **Boundary Type 1: Convergent:** Plates collide. The *heavy oceanic plate subducts* (sinks), while the *light continental plate moves up*, forming mountains! 🌰⬇️ *(Extracted from your detailed top-margin drawing!)*
- **Boundary Type 2: Divergent:** Plates pull apart, which causes *volcanoes to erupt*. 🌋 *(Extracted from your handwritten note!)*
- **Boundary Type 3: Transform:** Plates *slide horizontally* past each other, causing *earthquakes*. 😳 *(Extracted from your handwritten note!)*

The "Why?" Section ❓

- **Why do fold mountains form during tectonic movement?** Because the sedimentary rocks lying between colliding plates are severely compressed, and this massive compression directly causes the folding of the rocks. 🗜️〰️

3. Forces Causing Landforms 📐🌪️

Definitions:

- **Diastrophic Forces:** Forces originating deep inside the earth (Endogenic) that act horizontally and vertically to produce major relief like mountains, plateaus, plains, lakes, and big faults. 📐
- **Orogenic Movements:** Forces responsible for fold mountains, acting tangentially (horizontally) to the earth's surface. 🌰 ('Orogenic' = Greek for 'mountain-building'). They cause *severe deformation*. 💥 *(Handwritten note included!)*
- **Epeirogenic Movements:** Forces acting along the earth's radius (vertically) causing *large-scale* uplift or submergence of land areas. ⬆️⬇️ ('Epeiros' = continent, 'genic' = building). *(Handwritten "large scale" and "uplift ↑" notes included!)*

- **Exogenic Forces:** Forces operating entirely on the surface of the earth. 🌧️

Key Points:

- Epeirogenic movements result from faulting or cracks in the surface, producing block mountains, basins, and rift valleys on a very large scale. 🪨
- Exogenic forces are described as "Destructive Forces" because they cause widespread destruction through weathering and erosion, eventually forming new landforms through erosion and deposition. 🌪️

Table of Differences 🆚

Feature	Endogenic / Diastrophic Forces 🌋	Exogenic Forces 🌪️
Origin Point	Deep inside the earth.	Strictly on the earth's surface.
Primary Action	Constructive (uplift, folding, continent building).	Destructive (weathering and erosion).

4. Mountains & Fold Mountains ⛰️〰️

Definitions:

- **Mountain:** A very steep land or hill rising to great heights above the surrounding land. 📈
- **Fold Mountains:** Mountains resulting from large-scale lateral compression and stresses in the earth's crust. 〰️
- **Anticlines (Crest):** The upfold rock strata forming an arch-like shape. ∩
- **Synclines (Troughs):** The down-folded rock structures. ∪

Key Points (Exhaustive Detail & Critical Examples!):

- Mountains are classified into three types based on forming forces: Fold, Residual, and Block. 🔢

- Fold mountain stresses are caused by the weight of overlying rocks, mantle movements, or earth expansion/contraction, producing wrinkles along lines of weakness. 💥
- **CRITICAL EXAMPLES OF ARC-SHAPED MOUNTAIN CHAINS:** You will find arc-shaped chains globally, including the Andes (South America), Rockies (North America), Alps (Europe), Atlas (Africa), and the Himalayas, Zagros, Elburg, and Kunlun (Asia). 📌🌍
- **GLACIERS & PERENNIAL RIVERS:** Almost all high ranges (like the Himalayas and Alps) have peaks covered with snow throughout the year, hosting many glaciers. ❄️ These glaciers become the direct source of water for *perennial rivers*, making these mountains active agents of denudation and glacial erosion! 💧⛏️
- **OCEANIC FEATURES:** Fold chains beneath the sea have their tops revealed as *Islands* (e.g., West Indies). 🏝️ *Oceanic trenches* are formed by the exact same forces. 🌊 These forces combined with volcanic systems form *ridges* (e.g., the Mid-Atlantic Ridge, which rises 3 km above the ocean floor and joins the Indian Ocean Ridge). 🌋

Things to Remember: The 6 Chief Characteristics of Fold Mountains 🧠

Characteristic #	**Detailed Fact**
(i) Shape	They form irregular wave-like formations (folds) with Anticlines (crests) and Synclines (troughs). 〰️
(ii) Composition	Constituted of sedimentary rocks formed by deposition in shallow oceans (like the Tethys for the Himalayas). 🪨
(iii) Dimensions	They have greater length but are comparatively smaller in width (Himalayas are about 2400 km in length). 📏

Characteristic #	Detailed Fact
(iv) Location Rule	Most (except the Himalayas) are found on continent margins facing oceans. Rockies/Andes run North-South; Himalayas run West-East. 🧭
(v) Volcanism	Strongly associated with volcanic activity and possess many active volcanoes. 🌋
(vi) Plateaus	They have parallel ranges, and the high plateaus trapped between them are called *intermontane plateaus* (e.g., Anatolian Plateau in Turkey, Tibetan Plateau). ⛰️➖⛰️

1. How Oceanic Trenches Form (The Sinking Crust) 🕳️🌊

You know that when two plates push together (compression), they crumple upward to form fold mountains. But what happens when a **heavy oceanic plate** crashes into a **lighter continental plate?**

- Because the oceanic plate is so heavy, it doesn't crumple upward. Instead, it gets shoved downward, diving deep underneath the lighter plate into the Earth's mantle. 📉
- The point where this heavy plate gets dragged down creates a massive, steep, V-shaped ditch on the ocean floor. **This deep ditch is the Oceanic Trench!**
- This is why the book says trenches are the result of the "same forces as fold mountains." It is the exact same pushing force (compression), but because the oceanic crust is so heavy, it sinks to make a trench instead of rising to make a mountain! 💥

2. Underwater Fold Mountains & Islands 🏝️⛰️

Sometimes, when two plates crash underwater, they *do* crumple and push upward, creating a massive fold mountain range completely completely underwater.

- Over millions of years, these underwater mountains get pushed higher and higher.
- Eventually, the very top peaks of these mountains poke out above the surface of the ocean. To us standing above the water, these exposed mountain peaks look like **Islands** (like the West Indies mentioned in your book)! 🌴

3. Volcanic Systems & Ridges (The Pulling Apart) 🌋➡️⬅️

While trenches are formed by plates crashing together, **ridges** (like the Mid-Atlantic Ridge) are formed when plates do the opposite—they pull apart!

- Deep underwater, tectonic plates slowly tear away from each other. 💔
- This leaves a crack in the Earth's crust. Super-hot liquid rock (magma) from the Earth's mantle instantly rushes up to fill the gap. 🔥
- As soon as this magma hits the freezing ocean water, it cools and hardens into rock. As this happens over and over, it builds a massive, continuous chain of underwater volcanic mountains. This long, raised scar on the ocean floor is called a **Ridge**! ⛰️💧

5. Residual Mountains & Mountains of Denudation ⛰️🌬️

Definitions:

- **Residual / Relict Mountains:** Mountains which are subjected to weathering and erosion for a long time and are lowered down. 📉
- **Mountains of Denudation:** Another name for residual mountains, referring to the process of wearing away the earth's surface. 🌪️

Key Points (Exhaustive Detail):

- Denudation and erosion by natural agents (such as winds, water, snow, etc.) cause the mountains to be lowered, but some resistant areas may remain and form these residual mountains. 🪨
- They may also be formed on plateaus that are heavily dissected by rivers (e.g., the hills and valleys in the Deccan plateau). 🏞️
- The original highland (plateau) is actively worn down by the down-cutting action of rivers to form these residual mountains. 🌊✂️

- Most residual mountains are characterised by low elevations.
- However, both gentle and precipitous relief can exist, depending strictly on the degree of recent erosion. 📉
- **CRITICAL EXAMPLES:** Nilgiri, Parasnath, and Rajmahal Hills in India, and the Catskill range in the USA. 📌 INUS

The "Why?" Section ❓

- **Why do some parts remain tall while others are eroded?** Because the mountains are made of different rock types; the softer rocks are eroded away by agents like wind and water, while the highly *resistant areas* remain standing to form the residual mountains. 🛡️

Table of Differences (Young vs. Old Fold Mountains) 🆚

Feature	**Young Fold Mountains** ⛰️	**Old Fold Mountains** ⛰️
1. Time of Formation	They have been formed later than the Old Fold Mountains. *(Tick mark noted!)* ✔️	They have been formed long ago much before the Young Fold Mountains.
2. Examples	The Himalayas, Andes, Alps and Rockies are examples of young fold mountains.	The Appalachians, the Urals and the Aravallis (India) are examples of old fold mountains. *(Tick mark noted!)* ✔️
3. Physical Features	They have steep slopes, deep valleys and sharp pointed snow covered peaks.	They are highly eroded by the agents of gradation. They have gentler slopes, rounded tops and sculptured domes.

Feature	Young Fold Mountains ⛰️	Old Fold Mountains ⛰️
4. Height & Topography	They are much higher than the old fold mountains and have rough topography.	They are lower than the young fold mountains. *(Tick mark noted!)* ✔️
5. Volcanic Activity	Volcanic activities are common in these mountains.	There are evidences of volcanic activities still in them. *(Tick mark noted!)* ✔️

6. Block Mountains, Horsts & Rift Valleys 🪨↕️

Definitions:

- **Block Mountains:** Mountains formed strictly by the forces of faulting (also known as epeirogenic forces). 🧱
- **Horst Mountains:** Block mountains which have perfectly flattened summits. ➖
- **Rift Valley / Graben:** A subsided land or intervening basin formed when the land sinks between two parallel faults. 🕳️

Key Points (Exhaustive Detail & Handwritten Notes!):

- Due to tensional force, cracks or faults may occur on the earth's crust. ⚡
- If two parallel faults occur, the land in between subsides to form a rift valley. ⬇️
- If a block rises, it forms a horst or block mountain. ⬆️
- Block mountains that have tilted summits have one side steep, while the other side is very gentle. 📐
- Block mountains are very often separated by intervening basins called graben. 🏞️

- If there are two parallel faults, the valley floor may sink and a rift valley may be formed.
- **Handwritten Diagram Mastery:** In the formation of block mountains by tension, the raised blocks create a **"steep"** edge, while the land in between takes a drastic **"fall"** to create the Rift Valley (Graben). ✍️📉
- **CRITICAL EXAMPLES OF BLOCK MOUNTAINS:** Vosges (France), Black Forest mountains in Europe, Salt Range of Pakistan, and Sierra Nevada mountains of California, USA. 📌🌍
- **CRITICAL EXAMPLE OF RIFT VALLEY:** The most extensive system is the African Rift Valley. 📌🌍

The "Why?" Section ❓

- **Why do block mountains and rift valleys form?** They form due to *tensional forces* pulling the crust apart, causing cracks and faults. The earth's crust then shifts vertically (epeirogenic forces), causing blocks to either rise (Horst) or fall (Graben). ↔️↕️

7. Plateaus & Tablelands 🏜️

Definitions:

- **Plateau:** Defined as an area of high ground, 500 feet above the ground level. 📏
- **Tableland:** The upper part of a plateau which is generally completely flat. 🍽️

Key Points (Exhaustive Detail):

- These uplands have a steep slope on at least one side. 📉
- Plateaus are classified in several ways. Just as we used the criteria of "mode of formation" for mountains, we use the exact same criteria to classify plateaus. 📐

8. Intermontane Plateaus ⛰️➖⛰️

Definitions:

- **Intermontane Plateaus:** Plateaus surrounded by hills and mountains on all sides. ⛰️🏜️⛰️

Key Points (Exhaustive Detail & Handwritten Notes!):

- These are the highest and most extensive types of plateaus on the globe. 🌍
- They are formed directly along with fold mountains. 〰️
- **Handwritten Mastery:** Their formation is strongly associated with **"-> asymmetrical folding"** where the tectonic forces push unevenly, trapping the flat plateau between the rising mountain folds! ✍️ 💡
- **CRITICAL EXAMPLES:** The Tibetan plateau is the world's highest as well as most extensive intermontane plateau. Other examples are Columbian Plateau, North American Cordillera, Bolivian, Peruvian and Mexican plateaus. 📌 🗺️

9. Volcanic Plateaus 🌋 🧩

Definitions:

- **Volcanic Plateaus:** Plateaus formed by numerous volcanic eruptions that slowly build up over time. 🌋 ⏳

Key Points (Exhaustive Detail & Handwritten Notes!):

- **Handwritten Mastery (The Magma Rule):** These plateaus are formed specifically **"(when magma / basic less dense)"**. The basic, less viscous magma flows out smoothly over vast areas before cooling, creating a flat plateau instead of a steep mountain! ✍️ 💡
- The Plateau of Peninsular India, especially the north western Deccan, is a prime example of a lava plateau. IN
- The Peninsular plateau of India has an enormous accumulation of basaltic rock. 🪨
- The Columbian Plateau has about 20 layers of lava, which scientifically suggests that 20 completely different volcanic eruptions have formed this plateau. 🤩 🌋
- **CRITICAL EXAMPLES:** Plateau of Peninsular India (Deccan), South African plateau, Columbian Plateau, and the Ethiopian plateau (which you underlined!). 📌 🌍

Things to Remember 🧠

Feature	Detail
Plateau Minimum Height	500 feet above ground level. 📏
Most Extensive Rift Valley	African Rift Valley. 🌍
Highest & Most Extensive Plateau	Tibetan Plateau. 🏔️
Columbian Plateau Eruptions	Built from 20 different layers of lava! 🥞 🌋

10. Plains & Structural Plains 🌾 📐

Definitions:

- **Plain:** An extensive area of lowland with a level or gently undulating surface. 🏞️
- **Structural Plains:** The plains that are formed strictly due to the uplift or subsidence of land. ⬆️ ⬇️

Key Points (Exhaustive Detail & Handwritten Notes!):

- The formation of plains is one of the most intriguing landform development processes on the surface of the earth. There are seemingly endless variations in plains. 🌍
- They are an imprint of past and present processes. For example, a sudden heavy downpour may wash away a standing landform or fill up a depression to form a plain. 🌧️
- A plain is seldom formed by a single process. However, three main types can be distinguished: Structural Plains, Erosional Plains, and Depositional Plains. 🔢

- Diastrophic forces may cause the uplift of a portion of land beneath the *ocean water or may* cause submergence of coastal land under ocean water. *(Underlined in your book!)* 🌊⬆️
- The Great Plains of the USA are surrounded by Mississippi-Missouri Plains in the east and Rockies in the west. US
- On the other hand, the Coromandel plains in India are the result of mild subsidence followed by sedimentation. Thus, the forces of *structural development as well as deposition* were both active in the formation of these specific plains. IN

The "Why?" Section ❓

- **Why are there so many lakes in the Great Plains of the USA?** Because the Great Plains were formed due to the *uplift of land submerged under water*. The fact that they were previously submerged under water is directly responsible for the *existence of many lakes in the vast lowland*! 💧📍 *(Heavily circled and underlined in your notes!)*

  Understanding the Earth's Hidden Engine 🌍⚙️

  To understand how massive plains are created, we have to look deep underground. The Earth's surface looks solid, but it is actually riding on top of a highly active, moving layer of hot rock! Let's break down exactly how this works step-by-step. 🌋✨

  1. The Engine: Convection Currents 🔥🔄

  Deep inside the Earth (in a layer called the mantle), it is incredibly hot. This heat acts just like a pot of thick soup boiling on a stove.

- **The Rise:** The super-hot, semi-melted rock near the Earth's core becomes lighter and slowly rises toward the crust. 🔥⬆️
- **The Fall:** When it reaches just beneath the solid crust, it spreads out, cools down, becomes heavier, and sinks back down toward the core. ❄️⬇️
- This continuous, circular boiling motion is called a **convection current**. These currents are incredibly powerful and act like giant conveyor belts, pushing, pulling, and dragging the tectonic plates (the pieces of Earth's crust) above them. 🧩💪

2. Uplift (Emergence): Pushing the Land Up ⬆️🏔️

How it happens: When a massive convection current of hot mantle rock rises directly underneath a tectonic plate, it pushes upward against the solid crust. Think of it like a giant hand pushing up from underneath a rug. This immense pressure causes the land above to warp, bulge, and slowly lift higher in elevation. ✋🌎

Example: The Great Plains of the USA us 🌾

- **The Past:** Millions of years ago, the middle of North America was entirely underwater, covered by a shallow inland sea. 🌊
- **The Push:** Deep tectonic forces and rising convection currents began pushing the North American plate upward.
- **The Result:** As the land was slowly pushed up higher and higher (uplift), the ocean water naturally drained away, exposing the flat, muddy seabed to the sun. This massive dried-out seabed became the **Great Plains**. 🏜️
- **The Proof:** Because this land was an uneven ocean floor for millions of years, water gathered in the shallow dips left behind as it rose. This is why there are so many lakes in this vast, flat lowland today! 💧

3. Subsidence: Sinking the Land Down ⬇️📉

How it happens: On the other hand, where those convection currents cool and sink back down deep into the Earth, they can literally drag the crust downward with them. Sometimes, tectonic forces also stretch the crust out, making it thin and causing it to sag. This sinking motion is called **subsidence**. ⚓

4. Mild Subsidence + Sedimentation: The Flat Fill-In 🏖️🌊

How it happens: Sometimes the land doesn't sink drastically into a giant canyon; it just sinks a little bit near the edge of the ocean. This is called *mild subsidence*.

- When a coastal area sinks slightly, it creates a shallow, wide depression or "bowl." 🥣
- Rivers flowing down from nearby mountains carry tons of mud, silt, and sand (sediments). 🏞️

- When these fast rivers hit this mildly sunken, flat coastal area, they slow down and drop all their dirt into the depression. 🪨

  Example: The Coromandel Plains of India IN 🌴

- **The Sinking:** The land along the Coromandel coast experienced this mild subsidence, sinking slightly due to tectonic forces. 📉
- **The Filling:** Over thousands of years, rivers kept dumping massive amounts of sediment into this sunken area.
- **The Result:** The sediment acted like cement filling a pothole. It filled up the sunken land until it was completely flat, even, and level, creating a brand new, highly fertile structural plain! 🌾✨

  1. What is Mild Subsidence and How Does it Happen on the Coromandel Coast? 🌊IN

  To understand **mild subsidence**, let's contrast it with heavy, dramatic sinking. While a massive drop creates a deep ocean basin or trench, *mild subsidence* is a very gentle, subtle sagging or lowering of a land edge where the continent meets the sea.

- **The Tectonic Reason:** Deep underground, cooling convection currents or shifting tectonic weights can cause a coastal edge to lose its balance and tip slightly downward.
- **The "Mild" Part:** The land doesn't plunge thousands of feet into a dark abyss. Instead, it just dips a little bit—often only a few meters or tens of meters—causing the shallow edge of the sea to creep slightly inland over the low-lying coastal fringe. 📉
- **Why Sedimentation Follows Immediately:** Once this land sags into a shallow bowl, it creates a trap. Rivers flowing down from the high inland areas (like the Eastern Ghats) carry massive amounts of eroded rock, silt, and sand. When these rivers hit this slightly sunken, flat coastal zone, they slow down completely and drop their cargo.
- **The Final Result:** The sediment acts like a filler, neatly covering up the sunken depression and building a wide, perfectly flat, fertile plain right along the coast. That combination of *slight sinking* followed by *river-and-sea filling* is why the Coromandel coast exists! 🌾✨

2. What is an "Emerged Coastal Plain"? (The Gulf Coast Example) us 🌴

An **emerged coastal plain** is the exact opposite of a submerged or sinking coast. The word "emerged" means **to come out into view** or **rise up**.

- **How it Occurs:** Imagine a shallow part of the ocean right next to a continent. Because of tectonic uplift (those upward-pushing convection currents we talked about) or a drop in global sea levels, the seafloor is nudged upward just enough to peek above the water line. ⬆️🌊
- **The Gulf Coast Example:** The Gulf Coast region of the USA is a classic example of this. Parts of this land were previously sitting quietly under shallow sea waters. When tectonic forces gently lifted the edge of the continent (or when the water receded), that old seabed emerged into the open air as dry, flat land.
- **Why It Looks the Way It Does:** Because it used to be the bottom of the sea, an emerged coastal plain is typically very flat, sandy, and smooth, often featuring old marine fossils and low-lying flat terrain right next to the modern ocean. 🏖️

Things to Remember 🧠

Plain Type	Critical Examples 📌
Uplifted Submerged Land	The Great Plains of the USA.
Emerged Coastal Plain	Gulf Coast. *(Circled in your book!)*
Subsidence + Sedimentation	Coromandel plains in India.

11. Depositional Plains 🌊🏂

Definitions:

- **Depositional Plains:** The plains that are formed strictly by the deposition of materials which have been brought by various agents of transportation (rivers, wind, marine). 🚚🪨

- **Piedmont Alluvial Plains:** Plains formed exactly at the foothills of mountains. 🏔️
- **Polder:** A piece of land reclaimed from the sea or lake. The submerged land is surrounded by an embankment (dyke) and drained by pumping water into canals. 🚧💧

Key Points (Exhaustive Detail & Handwritten Notes!):

- **(i) River Deposition:** The most widespread of these are alluvial plains (like the Northern Plains of India), flood plains (like the Mississippi Plains in the USA), and deltaic plains (in Egypt and India). 🌊
- **Handwritten Mastery (The Piedmont Equation):** You perfectly broke down how Piedmont plains form! Here is your exact flow: **Piedmont -> River erode mountains -> water slows at foothills -> Heavy debris drop** ✍️💡
- These river plains are found along rivers like the Ganga, Yang-tze, Mekong, Salveen etc., and are regarded as the *"granaries of the world"*. 🌾
- **(ii) Wind Deposition:** Most such plains are sandy deserts like the Sahara in Africa and the Thar desert in India. They have an irregular and undulating surface made by the existence of sand dunes and hallows. 🐪 *(Checkmarks noted!)*
- Loess plains in China are formed directly from deserts via wind deposition. CN *(Checkmark noted!)*
- **(iii) Marine Deposition:** These plains are found in coastal regions. Examples include the east coast of India, the estuarine banks in the Netherlands, and the coasts of Germany and Denmark. 🌊 *(Checkmarks noted!)*

  MARINE DEPOSITION & RECLAIMED LAND 🏖️📝

  1. Natural Marine Deposition 🌊

- These plains are found in coastal regions.

- They are formed naturally when the ocean continuously deposits silt, sand, and mud onto the shore over thousands of years, building up flat land. 🏝️
- Examples are the east coast of India, the estuarine banks in the Netherlands, coast of Germany and Denmark.

2. Human-Made Land: Polders and Dykes 🧱🚜

- In Denmark, we find polders and dykes as a human-made shortcut to create land.
- **The Location:** The process starts past the coastline, right in the shallow ocean water. 🗺️
- **The Wall (Dyke):** An embankment (dyke) is built in a circle around that shallow ocean portion to block the sea.
- **The Drying:** The submerged land is then drained by pumping the trapped ocean water into canals.
- **The Result (Polder):** A polder is created, which is a piece of land reclaimed from the sea or lake. It becomes dry, usable land completely surrounded and protected by the embankment. 🛡️🏡

3. Why the Netherlands and Denmark? 🤔🌎 These areas are often called "low countries" because much of their geography sits naturally below sea level. To survive the threat of flooding and to create more living and farming space for a growing population, they utilize the shallow coastal waters. Instead of waiting for natural marine deposition, they build embankments to push the sea back and manually reclaim the ocean floor to build their own land. 🚜🌾

Table of Differences (Indian Flood Plains) 🆚

Feature	**Khadar Plains** 🌊	**Bhangar Plains** 🪨
Deposit Age	New deposits.	Old deposits.

Feature	Khadar Plains 🌊	Bhangar Plains 🪨
Renewal Rate	Made and renewed every single year.	Not renewed every year.

12. Importance of Landforms 🌍✨

Key Points (Exhaustive Detail & Handwritten Notes!):

- The landforms on the surface of the earth are symbols of the energy of the earth. ⚡
- They are basically a result of enormous heat generated inside the earth AND climate factors operating on the surface of the earth. 🔥☁️
- The earth releases its energy through tectonic activity, earthquakes, and volcanoes. 🌋
- These internal forces, in addition to surface weathering and erosional forces, become the dominant factors in the sculpturing of land. ⛏️
- **Advantage 1:** Mountains, plateaus, plains, and valleys are beautiful sculptures of nature on the surface crust. They add to the beauty of the earth. 🏞️
- **Advantage 2 (Climate Impact):** Landforms represent the relief features of the earth. They have a massive impact on climate, direction of winds, precipitation, and atmospheric processes.
    - *Critical Example:* The *Himalayas act as a climatic divide* shaping the distinctive climate of the Indian subcontinent. 🏔️☁️ *(Underlined and checked!)*
- **Advantage 3 (Water Source):** The mountain ranges of the world are the abode of snow and glaciers. They are thus the *source of perennial rivers.* ❄️💧 *(Underlined and checked!)*
- **Advantage 4 (Biodiversity):** Many kinds of forests are found on the slopes of mountains, hills, and plateaus. These are home to different species of plants and animals. Thus, the biodiversity of life on earth can be directly attributed to relief. 🌲🐆 *(Checked!)*

- **Advantage 5 (Fishing):** Since such relief features also exist in oceans, the island groups and coastal regions are *ideal fishing sites.* 🎣🏝️ *(Underlined and checked!)*
- **Advantage 6 (Resources):** Major landforms are created by disturbances resulting from internal earth forces. Because of this, most *mineral and fossil fuels come near the surface* of the earth due to these disturbances! 🪨⛽ *(Underlined and checked!)*

1. Identify the forces responsible for the formation of fold mountains.

- **Answer:** (a) Orogenic movements ⛰️

2. The large scale uplift or subsidence of land:

- **Answer:** (c) Epeirogenic movements ⬆️⬇️

3. The two forces, Orogenic and Epeirogenic that originate deep inside the earth:

- **Answer:** (c) Diastrophic forces 🌋

4. The forces operating on the surface of the earth which cause formation of landforms are called:

- **Answer:** (a) Exogenic forces 🌪️

5. Identify A and B. *(Image shows A pointing to the upfold/crest and B pointing to the downfold/trough)*

- **Answer:** (c) A: Anticline B: Syncline 〰️

6. Oceanic trenches in combination with volcanic mountain systems lead to the formation of:

- **Answer:** (b) Ridges 🌊⛏️

7. Volcanic activities are common in which of the following?

- **Answer:** (a) Young fold mountains 🌋

8. Assertion (A): External forces or Exogenic forces are Destructive Forces. Reason (R): They produce relief such as mountains, plateaus, plains, lakes and big faults.

- **Answer:** (c) A is true but R is false. *(Reason: Relief features like mountains are produced by internal/diastrophic forces, not exogenic forces!)* ⚖️

9. Assertion (A): Residual mountains are also called mountains of denudation. Reason (R): Denudation and erosion by natural agents cause mountains to be lowered.

- **Answer:** (a) Both A and R are true and R is the correct explanation of A. 📉

10. Which amongst the following highlands are highly eroded by the agents of gradation and have gentler slopes, rounded tops and sculptured domes?

- **Answer:** (b) Old fold mountains 🌰

11. Mountains which are subjected to weathering and erosion for a long time are lowered down:

- **Answer:** (c) Relict mountains 🪨

12. Which of the following is/are correctly matched?

- **Answer:** (d) All of the above. ✔️

13. Assertion (A): The forces of faulting lead to the formation of block mountains. Reason (R): Block mountains are often separated by intervening basins called graben.

- **Answer:** (b) Both A and R are true but R does not explain A. *(Reason: While both are facts, the presence of grabens is a result of the same forces, not the specific explanation for why faulting creates the mountains themselves.)* 🧱

14. The Coromandel plains along the east coast of India were formed as a result of two different activities. Name them.

- **Answer:** (a) Subsidence, deposition ⬇️🚚

15. The plains that are formed by the deposition of materials which have been brought by various agents of transportation:

- **Answer:** (b) Depositional plain 🌾

16. Depositional plains formed as a result of action of wind have irregular and undulating surface because of

- **Answer:** (b) existence of sand dunes and hallows. 🐫

17. A piece of land reclaimed from the sea or lake:

- **Answer:** (d) Polder 🚧

18. The plains that are formed due to uplift or subsidence of land are called:

- **Answer:** (a) Structural plains 📐

19. Which of the following pairs is incorrectly matched?

- **Answer:** (d) Marine Deposition Plains : Western Coastal Plains (India) *(Reason: Our notes specify the East coast of India for marine deposition examples!)* ❌

20. Epeirogenic forces lead to the formation of Block Mountains. In this context which of the following are true?

- **Answer:** (d) All of the above. 🪨

21. Name the plateaus, formed along with fold mountains, that are surrounded by hills and mountains on all sides.

- **Answer:** (c) Intermontane Plateaus ⛰️ ➖ ⛰️

22. Assertion (A): Many lakes exist in the vast lowland of the Great Plains of the USA. Reason (R): The Great Plains were formed due to uplift of land submerged under water.

- **Answer:** (a) Both A and R are true and R is the correct explanation of A. 💧📍

II. Short Answer Questions ✍️🌎

1. What is meant by a landform? Give two examples.

- **Answer:** A landform is a natural feature of the earth's surface. Two examples are mountains and plateaus. 🗻⛰️

2. Why are the fold mountains called so?

- **Answer:** They are called fold mountains because they are formed by severe lateral compression and stresses in the earth's crust, which causes the sedimentary rocks to wrinkle and compress into irregular wave-like formations known as folds (anticlines and synclines). 〰️🗜️

3. Give two chief characteristics of the fold mountains.

- **Answer:** (1) They form irregular wave-like formations with crests (anticlines) and troughs (synclines). (2) They are constituted of sedimentary rocks formed by deposition in shallow oceans. 🌊🪨

4. Give one example of each: (a) Young fold mountain; (b) Old fold mountain.

- **Answer:** (a) Young fold mountain: The Himalayas. 🏔️ (b) Old fold mountain: The Aravallis (India). 🏔️

5. Give one example of residual mountain and an example of depositional plain.

- **Answer:** Residual mountain: Nilgiri Hills. 🏔️ Depositional plain: Northern Plains of India (River Deposition). 🌾

6. Briefly describe how the Great Plains of USA were formed. They are examples of which type of Plains.

- **Answer:** The Great Plains of the USA were formed strictly due to the uplift of land that was previously submerged under water (which is why many lakes exist there). They are an example of **Structural Plains**. ⬆️💧

7. Draw a well labelled diagram to show the formation of a residual mountain.

- **Answer:** *(To draw this 🖍️: Draw a high, straight dotted line across the top and label it "Original surface" (a plateau). Below it, draw a wavy, uneven solid line showing deep valleys cut by rivers. Label this solid line "Surface as a result of denudation". Point arrows down from the dotted line to the wavy line to show erosion lowering the mountain!)* 📉✂️

8. How is a rift valley formed? Give one example of a rift valley.

- **Answer:** A rift valley is formed when tensional forces pull the crust apart, causing cracks or faults. If two parallel faults occur, the valley

floor (the land in between) subsides or sinks downward, creating a rift valley (graben). Example: The African Rift Valley. 🕳️⬇️

9. Give a brief definition of a plateau.

- **Answer:** A plateau is an area of high ground, at least 500 feet above ground level, that has a steep slope on at least one side and a generally flat upper part known as a tableland. 📏🍽️

10. What is an intermontane plateau? Give one example.

- **Answer:** An intermontane plateau is a high, extensive plateau that is surrounded by hills and mountains on all sides, formed along with fold mountains due to asymmetrical folding. Example: The Tibetan Plateau. ⛰️➖⛰️

11. Give two points of importance of landforms.

- **Answer:** (1) **Climate Impact:** They deeply impact climate, winds, and precipitation (e.g., the Himalayas act as a climatic divide). ☁️ (2) **Water Source:** Mountain ranges hold snow and glaciers, making them the direct source of perennial rivers. ❄️💧

Distinguish between each of the following: 🆚📊

1. Fold Mountains and Block Mountains.

- **Answer:**
    - **Fold Mountains:** Formed by horizontal, lateral compressive forces (Orogenic movements) that wrinkle and fold sedimentary rocks. (Example: Himalayas). 〰️
    - **Block Mountains:** Formed by vertical tensional forces and faulting (Epeirogenic movements) that cause blocks of the crust to rise or fall. (Example: Black Forest mountains). 🧱↕️

2. Intermontane Plateaus and Volcanic Plateaus.

- **Answer:**
    - **Intermontane Plateaus:** Formed alongside fold mountains due to asymmetrical folding, and are entirely surrounded by mountains. (Example: Tibetan Plateau). ⛰️

- **Volcanic Plateaus:** Formed by the buildup of multiple layers of basic, less-dense magma from numerous volcanic eruptions over time. (Example: Deccan Plateau). 🌋🤩

3. Structural Plains and Depositional Plains.

- **Answer:**
  - **Structural Plains:** Formed strictly by internal (diastrophic) forces causing the uplift or subsidence of land, such as submerged land rising. (Example: Great Plains of USA). ⬆️⬇️
  - **Depositional Plains:** Formed by external (exogenic) forces through the deposition of materials brought by agents of transportation like rivers, wind, or sea. (Example: Northern Plains of India). 🚚🪨

IV. Structured Questions 📝🌎

1. (a) How are various landforms formed on the surface of the earth?

- **Answer:** Landforms are formed by a combination of two major forces. **Endogenic (Diastrophic) forces** operate deep inside the earth due to intense heat and convection flows, causing plates to move, collide, or pull apart, which creates major relief like mountains and plateaus through uplift, folding, and faulting. Meanwhile, **Exogenic forces** operate on the surface, causing weathering, erosion, and deposition to further sculpt and create new landforms like plains and valleys. 🌋🌪️

1. (b) Describe the formation of mountains and their types.

- **Answer:** Mountains are formed mainly by tectonic forces and are classified into three types:
  1. **Fold Mountains:** Formed by horizontal, lateral compressive forces that severely compress sedimentary rocks, causing them to wrinkle and fold (e.g., Himalayas). 〰️
  2. **Block Mountains:** Formed by vertical tensional forces that create cracks or faults. The land blocks then rise (forming horsts) or fall due to epeirogenic movements. 🧱

3. **Residual Mountains:** Formed when agents of denudation (wind, water) erode original highlands over a long time, leaving only the hard, resistant rock areas standing. ✂️🏔️

1. (c) Give a reason for each of the following:

- **(i) Asthenosphere though underneath the crust is in a semi-molten state.**
    - **Answer:** The intense heat in the earth's interior (driven by radioactive decay and residual heat) melts the rocks, keeping the asthenosphere in a fragile, semi-molten/semi-magma state! 🔥🍯
- **(ii) Exogenic forces are also called destructive forces.**
    - **Answer:** They are called destructive because they cause widespread destruction of existing landforms through constant weathering and erosion before depositing the materials elsewhere. 🌪️💥
- **(iii) Residual mountains are also called mountains of denudation.**
    - **Answer:** Because they are formed strictly through the process of "denudation" and erosion, where natural agents wear down a plateau or highland, leaving only the most resistant rock behind. 🌧️🪨

1. (d) Draw a well labelled diagram to show the formation of block mountain.

- **Answer:** *(To draw this from our notes 🖍️: Draw three blocks of land. Draw arrows showing "Tension/Pull" pulling the outer blocks apart. Draw the middle block sliding downwards along crack lines (Faults). Label the dropped middle section "Rift Valley (Graben)" and the raised outer sections with flattened summits "Horst (Block Mountain)"!)* ↔️⬇️⬆️

2. (a) Describe the characteristics of two different types of plateaus.

- **Answer:**
    - **Intermontane Plateaus:** Formed alongside fold mountains due to asymmetrical folding, these are the highest and most extensive plateaus and are entirely surrounded by hills and mountains (e.g., Tibetan Plateau). 🏔️➖🏔️

- **Volcanic Plateaus:** Formed when less-dense, basic magma flows out from numerous volcanic eruptions over time. It builds up layer by layer into a flat highland rather than a steep cone (e.g., Deccan Plateau). 🌋🥞

2. (b) Describe the characteristics of structural plains.

- **Answer:** Structural plains are formed strictly due to the internal (diastrophic) uplift or subsidence of land. They bear the imprint of past and present tectonic processes. A prime characteristic is that they can form from land that was previously submerged under ocean water being uplifted (like the Great Plains of the USA) or from coastal submergence. ⬆️⬇️🌊

2. (c) Give a reason for each of the following:

- **(i) There are different types of plains.**
  - **Answer:** Because a plain is seldom formed by a single process! They are formed by seemingly endless variations and combinations of past and present processes, including uplift, subsidence, erosion, and multiple types of deposition. 🌍🔄
- **(ii) Alluvial plains are regarded as the granaries of the world.**
  - **Answer:** Because they are formed by river deposition (like the Ganga or Mississippi) which brings incredibly fertile sediments. Piedmont plains and flood plains renew these deposits, making the soil perfect for massive agricultural output. 🌾🚜
- **(iii) The landforms on the surface of the earth are symbols of the energy of the earth.**
  - **Answer:** Because they are basically the direct physical result of the enormous heat generated inside the earth, which releases its tectonic energy through earthquakes, volcanoes, and plate movements to sculpt the crust! ⚡🌋

2. (d) Explain the chief characteristics of depositional plains and their types.

- **Answer:** Depositional plains are formed when agents of transportation drop their carried materials. 🚚🪨
    - **River Deposition:** Forms widespread alluvial plains, flood plains, deltaic plains, and Piedmont plains at mountain foothills. 🌊
    - **Wind Deposition:** Forms sandy deserts with undulating surfaces, dunes, and hollows (like the Thar), as well as Loess plains. 🐫💨
    - **Marine Deposition:** Forms coastal plains and reclaimed land areas called Polders surrounded by dykes. 🏝️🚧

V. Thinking Skills 🧠💡

*(🚨 **OUTSIDE NOTES ALERT:** These questions require applying the textbook concepts to the real world! I have used our notes as the foundation and applied them to answer these intelligently for you!)*

1. Find out the landforms found in your state. How were these landforms formed? How many changes have come in these landforms over the years.

- **Answer:** In your state of Uttar Pradesh, the dominant landform is the massive **Depositional Plain**, specifically the **Northern Plains of India (Alluvial Plains)**! 🌾
    - **How they were formed:** As per our notes on *River Deposition*, these were formed by major rivers like the Ganga and Yamuna eroding the Himalayas and depositing massive amounts of highly fertile silt and debris as the water slowed down across the flat land. 🌊🟫
    - **Changes over the years:** These plains are constantly changing! As our notes mention, flood plains divide into *Khadar* (new deposits renewed every year by floods) and *Bhangar* (older, unrenewed deposits). Additionally, exogenic forces (and human agricultural activities) constantly reshape the river banks and surface soil! 🔄🚜

2. The earth releases its energy through tectonic activity, earthquakes and volcanoes. Imagine what will be the consequences if this energy remains trapped inside the earth. State these consequences with examples.

- **Answer:** If the enormous internal heat and energy from radioactive decay could not escape, the consequences would be catastrophic! 😱
    - **Consequence 1 (No New Landforms):** The convection currents would stop functioning properly. Without tectonic plate movement (diastrophic forces), no new fold mountains, rift valleys, or plateaus would ever form. Exogenic forces (wind/water) would eventually erode all existing landforms flat, turning the earth into a smooth, featureless sphere. 🌍📉
    - **Consequence 2 (Catastrophic Explosion):** Pressure would build up endlessly. Eventually, the trapped magma and gases would fracture the entire lithospheric crust at once, causing a planet-wide apocalyptic volcanic eruption, completely destroying the crust and all life on it, rather than releasing safely through smaller, regular volcanic plateaus or ridges! 💥🌋